Question ID: c048055c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question

A model predicts that the population of Bergen was $15{,}000$ in 2005. The model also predicts that each year for the next 5 years, the population p increased by 4% of the previous year's population. Which equation best represents this model, where x is the number of years after 2005, for $x \leq 5$?

Answer

A. $p = 0.96(15{,}000)^x$

B. $p = 1.04(15{,}000)^x$

C. $p = 15{,}000(0.96)^x$

D. $p = 15{,}000(1.04)^x$

Correct Answer: D

Rationale

Choice D is correct. It's given that a model predicts the population of Bergen in 2005 was $15{,}000$. The model also predicts that each year for the next 5 years, the population increased by 4% of the previous year's population. The predicted population in one of these years can be found by multiplying the predicted population from the previous year by 1.04. Since the predicted population in 2005 was $15{,}000$, the predicted population 1 year later is $15{,}000(1.04)$. The predicted population 2 years later is this value times 1.04, which is $15{,}000(1.04)(1.04)$, or $15{,}000(1.04)^2$. The predicted population 3 years later is this value times 1.04, or $15{,}000(1.04)^3$. More generally, the predicted population, p, x years after 2005 is represented by the equation $p = 15{,}000(1.04)^x$.

Choice A is incorrect. Substituting 0 for x in this equation indicates the predicted population in 2005 was 0.96 rather than $15{,}000$.

Choice B is incorrect. Substituting 0 for x in this equation indicates the predicted population in 2005 was 1.04 rather than $15{,}000$.

Choice C is incorrect. This equation indicates the predicted population is decreasing, rather than increasing, by 4% each year.